Math 105L: Using First and Second Derivatives II

Lecture 11-1 Completed Board Lecture & Discussion Sheet

Duke University | Department of Mathematics | Monday, November 2, 2026

1. Theoretical Foundation: Extrema vs. Inflection Points

First Derivative Test vs. Points of Inflection

- **Local Extrema (First Derivative Test)**: Let a be in the interior of the domain of continuous f. If $f'(x)$ changes sign across a, then $(a, f(a))$ is a local extremum. Specifically, a transition from positive to negative $(+ \to -)$ yields a **local maximum**, while negative to positive $(- \to +)$ yields a **local minimum**. A critical point ($f' = 0$ or undefined) is a *necessary candidate*, but not sufficient on its own.
- **Points of Inflection**: A point $(c, f(c))$ on the curve is an **inflection point** if f is continuous at c and the **concavity changes sign** across c (f'' transitions $+ \to -$ or $- \to +$). The algebraic condition $f''(c) = 0$ (or undefined) identifies *candidates*, but concavity must strictly alternate across the point.

Common Exam Fallacy: Setting $f''(x) = 0$ Does NOT Find Extrema!

Setting $f''(x) = 0$ identifies candidate inflection points, **not** local extrema. Local extrema occur exclusively at critical points ($f' = 0$ or undefined) where f' changes sign. The **Second Derivative Test** evaluates the sign of $f''(a)$ at an *already known* critical point where $f'(a) = 0$ ($f''(a) > 0 \implies$ local min; $f''(a) < 0 \implies$ local max; $f''(a) = 0 \implies$ inconclusive).

Comparative Benchmark: Why Candidates Must Be Tested for Sign Changes

Function	$f'(x)$	$f''(x)$	At $x = 0$	Geometric Classification at $x = 0$
$y = x^3$	$3x^2 \geq 0$	$6x$	$f'(0) = 0, f''(0) = 0$	**Inflection Point** (f'' changes sign $- \to +$). Neither local max nor min.
$y = x^4$	$4x^3$	$12x^2 \geq 0$	$f'(0) = 0, f''(0) = 0$	**Local Minimum** (f' changes $- \to +$). *NOT* an inflection point ($f'' \geq 0$).
$y = -x^4$	$-4x^3$	$-12x^2 \leq 0$	$f'(0) = 0, f''(0) = 0$	**Local Maximum** (f' changes $+ \to -$). *NOT* an inflection point ($f'' \leq 0$).

2. Case Study: High-Degree Polynomial $f(x) = (x^2 - 4)^7$

Global Algebraic Structure:

- **Degree & Parity**: Degree $2 \times 7 = 14$ (even degree, positive leading coefficient $\implies \lim_{x \to \pm\infty} f(x) = +\infty$). Even function ($f(-x) = f(x)$), reflecting perfect bilateral symmetry across the y-axis.
- **Roots (x-intercepts)**: Setting $f(x) = 0 \implies (x^2 - 4)^7 = 0 \implies x^2 - 4 = 0 \implies \mathbf{x = \pm 2}$.

First Derivative & Critical Point Classification: Applying the Generalized Power Rule (Chain Rule):

$$f'(x) = 7(x^2 - 4)^6 \cdot \frac{d}{dx}[x^2 - 4] = 7(x^2 - 4)^6 \cdot (2x) = \mathbf{14x(x^2 - 4)^6}$$

Setting $f'(x) = 0$ yields critical points at:

$$14x = 0 \implies \mathbf{x = 0}, \qquad \text{or} \qquad (x^2 - 4)^6 = 0 \implies x^2 = 4 \implies \mathbf{x = \pm 2}.$$

First Derivative Sign Mechanics: Because $(x^2 - 4)^6$ has an even exponent (6), it is **strictly non-negative** for all real numbers ($(x^2 - 4)^6 \geq 0$). Consequently, the sign of $f'(x)$ is governed entirely by the linear factor $14x$:

- On $(-\infty, 0)$: $x < 0 \implies f'(x) < 0$, so $f(x)$ is **strictly decreasing**.
- On $(0, \infty)$: $x > 0 \implies f'(x) > 0$, so $f(x)$ is **strictly increasing**.
- **At** $x = 0$: f' transitions from negative to positive $(- \to +) \implies$ **Strict Local Minimum** at $(0, (-4)^7) = (0, -16384)$.
- **At** $x = \pm 2$: f' attains zero but does **not** change sign ($f' \leq 0$ on $(-\infty, 0]$ and $f' \geq 0$ on $[0, \infty)$) $\implies$ **Horizontal shelf points (Neither local max nor min)**!

2. Case Study: High-Degree Polynomial (Continued)

Second Derivative Derivation via Product & Chain Rules: Differentiating $f'(x) = 14x \cdot (x^2-4)^6$:

$$\begin{aligned} f''(x) &= \frac{d}{dx}[14x] \cdot (x^2-4)^6 + 14x \cdot \frac{d}{dx}\left[(x^2-4)^6\right] \\ &= 14(x^2-4)^6 + 14x \cdot \left[6(x^2-4)^5 \cdot (2x)\right] \\ &= 14(x^2-4)^6 + 168x^2(x^2-4)^5 \end{aligned}$$

The "Never Expand High School Algebra" Factoring Rule

Expanding $(x^2-4)^6$ into a degree-12 polynomial is algebraically catastrophic. Instead, factor out the greatest common factor $\mathbf{14(x^2-4)^5}$ directly:

$$f''(x) = 14(x^2-4)^5\left[(x^2-4)+12x^2\right] = \mathbf{14(x^2-4)^5(13x^2-4)}$$

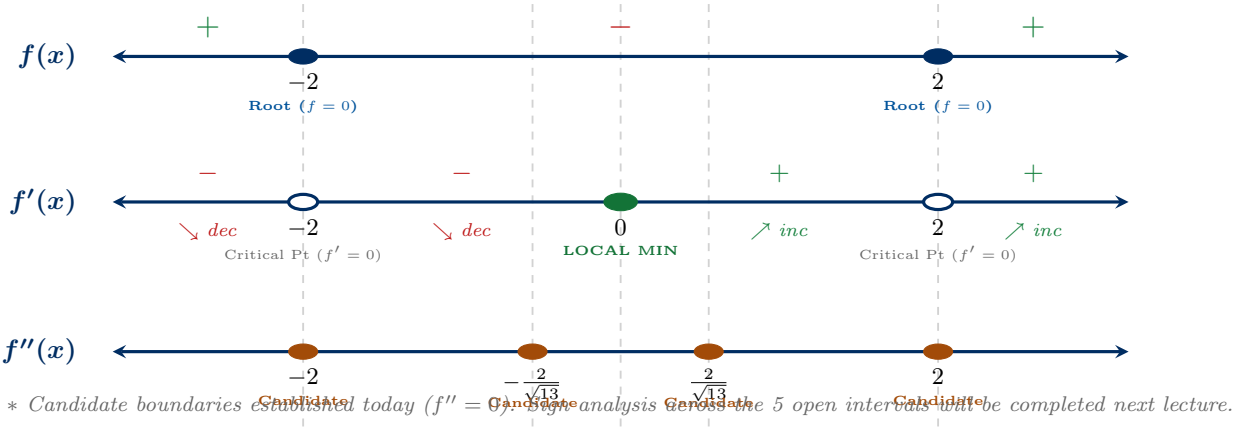
Candidate Inflection Points: Setting $f''(x) = 0$ yields two distinct factors to solve:

1. $(x^2-4)^5 = 0 \implies x^2-4=0 \implies \mathbf{x = \pm 2}$.
2. $13x^2-4=0 \implies x^2 = \frac{4}{13} \implies \mathbf{x = \pm\frac{2}{\sqrt{13}}} \approx \pm 0.5547$.

Because $\sqrt{13} \approx 3.606$, $\frac{2}{\sqrt{13}} < 2$. Thus, we obtain four candidate inflection points: $\mathbf{x = -2}$, $\mathbf{-\frac{2}{\sqrt{13}}}$, $\mathbf{\frac{2}{\sqrt{13}}}$, $\mathbf{2}$.

3. Aligned Comparative Number Lines: $f(x)$, $f'(x)$, and $f''(x)$

The diagram below connects key x-coordinates vertically across all three functions to display how roots, critical points, and candidate inflection points align geometrically.



∗ *Candidate boundaries established today ($f'' = 0$). Sign analysis across the 5 open intervals will be completed next lecture.*

Key Takeaways & Next Steps

- **Critical Points vs. Extrema**: $f'(x) = 14x(x^2-4)^6$ has three critical points ($x = 0, \pm 2$). Only $x = 0$ is an extremum (local minimum) because f' changes sign. At $x = \pm 2$, f' does not change sign, producing horizontal shelf/terrace points.
- **Factoring Strategy**: When computing higher derivatives of composite functions, always factor out the lowest power of the repeated factor ($14(x^2-4)^5$). Never expand!
- **Candidates for Inflection**: Setting $f''(x) = 0$ produces four candidates: $x = \pm 2$ and $x = \pm\frac{2}{\sqrt{13}}$. Testing whether concavity actually alternates across each candidate will be our starting point on Wednesday.